Data Dictionary for Data-Driven Media Optimization Project

Project Context

This data dictionary supports the capstone project "Data-Driven Media Optimization for E3 Media Crafters Pvt Ltd," which aims to enhance marketing strategies by analyzing customer feedback, predicting campaign performance trends, and classifying visual content using the Yelp Open Dataset. The project integrates Python, Deep Learning (NLP, TSA, CV), Power BI, and Streamlit, following the CRISP-DM framework within an 8-week internship. This dictionary provides learners with a clear understanding of the dataset's structure and its relevance to project needs, facilitating data preparation, modeling, and analysis.

Dataset Overview

The Yelp Open Dataset is a publicly available resource containing business, review, and photo data. For this project, we focus on a subset (e.g., restaurants in Las Vegas, 2018-2023) to manage computational demands while ensuring relevance to E3 Media Crafters' goals of optimizing media campaigns. The dataset is divided into three main components: reviews (for NLP), timestamps (for TSA), and photos (for CV).

Data Dictionary

The following tables describe the key files and fields used in the project, including their purpose, format, and relevance to specific tasks.

1. Reviews Data (reviews.json)

Field Name	Data Type	Description	Purpose
review_id	String	Unique identifier for each review	Links reviews to other data; used for tracking
business_id	String	Unique identifier for the business	Connects reviews to businesses and photos
text	String	Full text of the review	Input for NLP sentiment analysis
stars	Integer	Rating given by the user (1-5)	Used to derive sentiment labels (1-2: negative, 3: neutral, 4-5: posit
date	DateTime	Date and time of the review	Used for TSA to create time series

Purpose in Project:

- **NLP**: The `text` field is the primary input for sentiment analysis, where learners train models (e.g., BERT, LSTM) to classify reviews as positive, neutral, or negative.
- **TSA**: The `date` field, combined with `stars` or derived sentiment, supports time series aggregation (e.g., monthly average sentiment per business).

- **Relevance to E3**: Understanding customer sentiment helps E3 tailor campaigns to address feedback, enhancing engagement.

2. Business Data (business.json)

Field Name	Data Type	Description	Purpose	Exan
business_id	String	Unique identifier for the business	Links businesses to reviews and photos	"def456uvw(
name	String	Name of the business	Contextual information for analysis	"Joe's Diner"
city	String	City where the business is located	Filters data (e.g., Las Vegas subset)	"Las Vegas"
categories	String	Tags describing the business	Filters for relevant businesses (e.g., restaurants)	"Restaurants
attributes	Dictionary	Key-value pairs of business features	Labels for CV tasks (e.g., ambience_is_classy)	{"ambience_

Purpose in Project:

- **CV**: The `attributes` field provides labels like "ambience_is_classy" or "good_for_kids" for training CV models to classify photos.
- **Data Filtering**: Fields like `city` and `categories` help learners select a manageable subset (e.g., restaurants).
- **Relevance to E3**: Business attributes guide campaign visuals, ensuring alignment with audience preferences.

3. Photos Data (photos.json)

Field Name	Data Type	Description	Purpose	Example Value
photo_id	String	Unique identifier for each photo	Tracks images for CV tasks	"ghi789rst345"
business_id	String	Links photo to a business	Connects photos to attributes and reviews	"def456uvw012"
caption	String	Description of the photo	Contextual input for CV or NLP	"Cozy dining area"
label	String	Category of the photo	Guides CV task selection	"inside"
image_path	String	File path to the image	Accesses image for processing	"photos/ghi789rst345.j

Purpose in Project:

- **CV**: The `image_path` provides access to images, which learners pair with `attributes` from `business.json` (via `business_id`) to train CNNs for classification (e.g., predicting "ambience_is_classy").
- **NLP (Optional)**: The `caption` field can supplement sentiment analysis if needed.
- **Relevance to E3**: Classifying visual content helps E3 design appealing campaign imagery.

4. Derived Datasets (Created by Learners)

Dataset Name	Fields	Data Type	Description
Sentiment Data	review_id, preprocessed_text, sentiment_label	String, String, String	Cleaned reviews with sentiment lab
Time Series Data	date, business_id, avg_sentiment	DateTime, String, Float	Monthly sentiment scores per busir
Image Data	photo_id, image_path, attribute_label	String, String, Integer	Images with binary labels (e.g., 1 fo

Purpose in Project:

- These datasets are outputs of data preparation tasks, tailored for modeling:

    - **Sentiment Data**: Feeds into NLP models.
    - **Time Series Data**: Used for TSA models like LSTM.
    - **Image Data**: Supports CV tasks with labeled images.
- **Relevance to E3**: Structured data enables precise analysis, driving campaign optimization.

Usage Notes

- **Subset Selection**: To manage scale, learners should filter the dataset (e.g., restaurants in Las Vegas, 2018-2023) to reduce processing time while maintaining relevance.
- **Preprocessing**:
    - **NLP**: Remove stop words, tokenize text, handle missing values in `text`.
    - **TSA**: Aggregate `stars` or sentiment by month, ensuring no gaps in time series.
    - **CV**: Resize images to 224x224 pixels, normalize pixel values (0-1).
- **Joins**:
    - Link `reviews.json` and `photos.json` to `business.json` using `business_id`.
    - Example: Match photos to attributes for CV tasks.
- **Storage**: Store raw data as JSON, derived datasets as CSV for compatibility with Power BI and Streamlit.

Relevance to Project Needs

The Yelp Open Dataset aligns with E3 Media Crafters' goals:

- **Sentiment Analysis (NLP)**: Review text and stars enable E3 to gauge audience reactions, informing campaign tone.
- **Performance Forecasting (TSA)**: Timestamps allow prediction of engagement trends, optimizing resource allocation.
- **Image Classification (CV)**: Photos and attributes help E3 design visuals that resonate with target audiences.
- **Visualization (Power BI)**: Aggregated data supports dashboards for stakeholder insights.
- **Interactivity (Streamlit)**: Processed data powers a user-friendly app for campaign analysis.

Example Workflow

1. **Week 2 (Data Understanding)**: Learners explore `reviews.json` to plot star distributions, confirming data quality.
2. **Week 3 (Data Preparation)**: Clean `text` field, creating `Sentiment Data` with labels.
3. **Week 4**: Aggregate `stars` by month for `Time Series Data`, pair photos with attributes for `Image Data`.
4. **Week 5-7 (Modeling)**: Use datasets for NLP, TSA, and CV models.

5. **Week 8 (Deployment)**: Export datasets to Power BI for dashboards and Streamlit for app integration.

Notes for Learners

- **Access**: Download the dataset from Yelp Open Dataset.
- **Tools**: Use Python (Pandas, NumPy) for processing, TensorFlow/Keras for modeling.
- **Challenges**:
    - Handle missing captions in `photos.json` by relying on attributes.
    - Balance sentiment classes if skewed (e.g., more 4-5 star reviews).
    - Optimize image processing to avoid memory issues.
- **Support**: Refer to the project charter for task guidance and consult the PM during weekly check-ins.